

Paper 2 - Set 10 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

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3	D	Integration
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Worked solutions

Question 1 IND-04 · Indices and surds · spec MM1.2

Express $\frac{5}{3 + 2\sqrt{5}}$ in the form $a + b\sqrt{5}$ where a and b are rational.

Answer: B

Multiply top and bottom by $3 - 2\sqrt{5}$: denominator $9 - 20 = -11$, numerator $15 - 10\sqrt{5}$. So the value is $\frac{15 - 10\sqrt{5}}{-11} = \frac{-15 + 10\sqrt{5}}{11}$.

Full method - how and why

Rationalising a denominator of the form $p + q\sqrt{n}$. The tool is the difference of two squares, which turns the surd denominator into an integer.

1 Multiply numerator and denominator by the conjugate $3 - 2\sqrt{5}$.

why: $(p + q\sqrt{n})(p - q\sqrt{n}) = p^2 - q^2n$, which contains no surd. Multiplying top and bottom by the same thing does not change the value.

2 Denominator: $3^2 - (2\sqrt{5})^2 = 9 - 4 \times 5 = -11$.

why: Square the **whole** second term, coefficient included: $(2\sqrt{5})^2 = 4 \times 5$, not 2×5 .

3 Numerator: $5(3 - 2\sqrt{5}) = 15 - 10\sqrt{5}$.

why: Nothing subtle - just expand.

4 Tidy the signs: $\frac{15 - 10\sqrt{5}}{-11} = \frac{-15 + 10\sqrt{5}}{11}$.

why: A negative denominator is not 'in the form $a + b\sqrt{5}$ ' with the usual convention, and both versions are offered, so the sign handling decides the mark.

Where it goes wrong

- $(2\sqrt{5})^2 = 10$ (forgetting to square the 2) gives denominator -1 and a wildly different answer.
- Leaving $\frac{15 - 10\sqrt{5}}{-11}$ and picking the option with $+11$ in the denominator but the wrong signs on top.

Takeaway Multiply by the conjugate, square the coefficient as well as the surd, then fix the sign of the denominator at the end.

Question 2 LIN-14 · Straight lines · spec MM3.1

A line passes through $(1, 2)$ with gradient m . It crosses the x -axis at A and the y -axis at B , and the midpoint of AB is $(1, 2)$. Find m .

Answer: A

Midpoint $(1, 2)$ means $A = (2, 0)$ and $B = (0, 4)$. Gradient $= \frac{4-0}{0-2} = -2$.

Full method - how and why

The midpoint condition determines both intercepts, and two points give the gradient.

1 Let $A = (p, 0)$ and $B = (0, q)$. Their midpoint is $(\frac{p}{2}, \frac{q}{2})$.

why: One intercept on each axis, so each has a zero coordinate.

2 Set the midpoint equal to $(1, 2)$: $\frac{p}{2} = 1$ and $\frac{q}{2} = 2$, so $p = 2$, $q = 4$.

why: Equate coordinates separately.

3 Gradient through $(2, 0)$ and $(0, 4)$: $\frac{4-0}{0-2} = -2$.

why: Consistent order of subtraction in numerator and denominator.

Where it goes wrong

- Halving instead of doubling, giving $A = (\frac{1}{2}, 0)$ and the wrong gradient.
- Answering 2 by ignoring that the line falls from left to right.

Takeaway If $(1, 2)$ is the midpoint of the intercepts, the intercepts are $(2, 0)$ and $(0, 4)$ - double each coordinate.

Question 3 INT-07 · Integration · spec MM7.3

Given that $\int_0^a (2x + 1) dx = 12$ where $a > 0$, find a .

Answer: D

$$a^2 + a = 12 \Rightarrow (a + 4)(a - 3) = 0 \Rightarrow a = 3.$$

Full method - how and why

A definite integral with an unknown limit. Integrate, substitute, and solve the resulting equation.

- 1 Integrate: $\int_0^a (2x + 1) dx = [x^2 + x]_0^a = a^2 + a$.
why: The lower limit 0 contributes nothing, which keeps this clean.
- 2 Set equal to 12: $a^2 + a - 12 = 0$.
why: Bring everything to one side.
- 3 Factorise: $(a + 4)(a - 3) = 0$, so $a = -4$ or $a = 3$.
why: Two numbers multiplying to -12 and adding to 1.
- 4 Apply the condition $a > 0$: $a = 3$.
why: The question's restriction exists to select one root.

Where it goes wrong

- Forgetting the '+x' from integrating the constant 1, giving $a = \sqrt{12}$ - an offered option.
- Giving both roots when only the positive one is allowed.

Takeaway An unknown limit becomes an equation after integration; use any stated sign condition to choose the root.

Question 4 LGC-13 · The logic of arguments · spec Arg4

Which of the following is the negation of ' $x > 3$ and $y \leq 5$ '?

Answer: E

Not (A and B) is (not A) or (not B). Not $x > 3$ is $x \leq 3$; not $y \leq 5$ is $y > 5$.

Full method - how and why

De Morgan's law: negating an 'and' turns it into an 'or', with both parts negated.

- 1 $\neg(A \text{ and } B) \equiv (\neg A) \text{ or } (\neg B)$.
why: For the conjunction to fail, it is enough that **one** part fails.
- 2 Negate the first part: $\neg(x > 3)$ is $x \leq 3$.
why: The boundary $x = 3$ satisfies the negation, since $3 > 3$ is false.
- 3 Negate the second: $\neg(y \leq 5)$ is $y > 5$.
why: Strict inequality, because $y = 5$ satisfied the original.
- 4 Combine with 'or': $x \leq 3$ **or** $y > 5$.
why: The joining word flips too - this is the half most often forgotten.

Where it goes wrong

- Keeping 'and': ' $x \leq 3$ and $y > 5$ ' is far too strong - it demands both failures.
- Getting the boundaries wrong: writing $x < 3$ or $y \geq 5$, which mishandles $x = 3$ and $y = 5$.

Takeaway De Morgan: $\text{not}(A \text{ and } B) = \text{not}A \text{ or } \text{not}B$; $\text{not}(A \text{ or } B) = \text{not}A \text{ and } \text{not}B$. Negate the connective as well as the parts.

Question 5 RSN-07 · Chains of reasoning · spec Prf5

The digits 1 to 9 are placed in a 3×3 grid so that every row and every column has the same sum. What is that sum?

Answer: B

The three rows together contain every digit once, so $3 \times (\text{row sum}) = 1 + 2 + \dots + 9 = 45$, giving 15.

Full method - how and why

Count the same total two different ways - the standard double-counting move.

- 1 The three rows between them contain each digit 1 to 9 exactly once.
why: Every cell lies in exactly one row, so the rows partition the grid.
- 2 So the three row sums add to $1 + 2 + \dots + 9 = 45$.
why: Use $\frac{9 \times 10}{2} = 45$.
- 3 All three row sums are equal, so each is $\frac{45}{3} = 15$.
why: And the columns give the same total, so they must each be 15 too - consistent.

Where it goes wrong

- Answering 45, the grand total rather than one row's sum.
- Trying to construct an actual magic square; the total settles it with no construction needed.

Takeaway Double-count a total: rows partition the grid, so $(\text{number of rows}) \times (\text{row sum}) = \text{grand total}$.

Question 6 ERR-04 · Identifying errors in proofs · spec Err1

Claim: for all real x , if $x^2 > 9$ then $x > 3$.

- I Suppose $x^2 > 9$.
- II Taking square roots of both sides, $x > 3$.
- III This proves the claim.

Which of the following is true?

Answer: A

$x^2 > 9$ gives $|x| > 3$, i.e. $x > 3$ or $x < -3$. The claim itself is false, and line II is where the invalid deduction happens.

Full method - how and why

Taking square roots of an inequality gives a modulus, not a bare inequality. Here the claim itself is false.

- 1 Test the claim: $x = -4$ gives $x^2 = 16 > 9$, but $x = -4 > 3$. The **claim is false**.
why: Settle the truth of the claim first; that tells you an error must exist somewhere.
- 2 Line I is just the supposition - no error.
why: You are allowed to suppose the hypothesis.
- 3 Line II says 'taking square roots, $x > 3$ '. What $x^2 > 9$ actually gives is $|x| > 3$, i.e. $x > 3$ **or** $x < -3$. **First error**.
why: $\sqrt{x^2} = |x|$, not x .
- 4 The offered repair ('say $|x| > 3$ and then deduce $x > 3$ ') is no repair at all, since $|x| > 3$ does **not** give $x > 3$.
why: That option admits the modulus but then makes the same invalid step.

Where it goes wrong

- Accepting the option that says the claim is true - it is not.
- Blaming line III; by then the damage is done.
- Solving $x^2 > 9$ as $-3 < x < 3$, which is the solution of $x^2 < 9$.

Takeaway $x^2 > k^2 \Leftrightarrow |x| > k \Leftrightarrow x > k$ or $x < -k$. Squaring and rooting both lose sign information.

Question 7 LGC-01 · The logic of arguments · spec Arg1

Consider the statement: **if n is a multiple of 6, then n is a multiple of 3**. Which of the following is its contrapositive?

Answer: A

The contrapositive of 'if A then B ' is 'if not B then not A '. It is logically equivalent to the original; B is the converse and C is the inverse.

Full method - how and why

The four relatives of an implication. Given $A \Rightarrow B$, you must be able to write the converse, inverse and contrapositive without hesitating.

- 1 Name the parts: A is ' n is a multiple of 6', B is ' n is a multiple of 3'.
why: Always label before manipulating - most errors here are bookkeeping errors.
- 2 The contrapositive is 'if not B then not A ': **if n is not a multiple of 3, then n is not a multiple of 6**.
why: Negate both parts **and** swap them. Doing only one of the two gives the wrong relative.
- 3 Sanity-check the meaning: if 3 does not divide n , then 6 certainly cannot, since every multiple of 6 is a multiple of 3.
why: The contrapositive is logically **equivalent** to the original, so it must also be true - a free check.

Where it goes wrong

- Choosing 'if n is a multiple of 3 then n is a multiple of 6' - that is the **converse** (swap only), and it is false ($n = 9$).
- Choosing 'if n is not a multiple of 6 then n is not a multiple of 3' - the **inverse** (negate only), also false ($n = 9$).

Takeaway Converse: swap. Inverse: negate. Contrapositive: swap **and** negate - and only the contrapositive is equivalent to the original.

Question 8 X2-04 · Sequences and series · spec MM2.3

A geometric series has first term $a \neq 0$ and common ratio r . P : ' $-1 < r < 1$ '. Q : 'the series has a sum to infinity'. Which one of the following is true?

Answer: A

A geometric series converges if and only if $|r| < 1$, which is exactly $-1 < r < 1$.

Full method - how and why

A definition-driven case: the convergence condition for a geometric series is exactly the stated inequality, so both directions hold.

- 1 A geometric series with $a \neq 0$ has a sum to infinity if and only if $|r| < 1$.
why: The partial sums $\frac{a(1-r^n)}{1-r}$ converge precisely when $r^n \rightarrow 0$.
- 2 $|r| < 1$ is the same statement as $-1 < r < 1$.
why: Modulus inequalities unfold into a double inequality.
- 3 So P and Q are equivalent: necessary **and** sufficient.
why: An 'if and only if' condition gives both directions at once.

Where it goes wrong

- Forgetting the case $r = \pm 1$: $r = 1$ gives partial sums na , which diverge; $r = -1$ oscillates. Both are correctly excluded.
- Answering 'sufficient but not necessary' from habit - here no counterexample exists.
- Overlooking why $a \neq 0$ is stated: with $a = 0$ every r would give a (trivial) sum.

Takeaway Geometric series converge $\Leftrightarrow |r| < 1$, and then $S_\infty = \frac{a}{1-r}$. An 'if and only if' condition is necessary and sufficient.

Question 9 LGC-14 · The logic of arguments · spec Arg1

For a positive integer n , consider the statement: **if n is prime, then n is odd**. Which one of the following is true?

Answer: D

$n = 2$ is an even prime, so the statement is false. The converse, 'if n is odd then n is prime', fails for $n = 9$.

Full method - how and why

Statement and converse again, with the classic $n = 2$ counterexample.

- 1 Statement: primes are odd? $n = 2$ is prime and even. **False**.
why: 2 is the only even prime - it is the standard counterexample and worth memorising.
- 2 Converse: odd numbers are prime? $n = 9 = 3 \times 3$ is odd and composite. **False**.
why: Also $n = 1$, which is odd and not prime by definition.
- 3 So both are false.
why: Match to the option list.

Where it goes wrong

- Forgetting that 2 is prime and declaring the statement true.
- Forgetting that 1 is **not** prime when hunting counterexamples.

Takeaway 2 is the even prime and 1 is not prime - the two facts behind most prime-related counterexamples.

Question 10 RSN-22 · Chains of reasoning · spec Prf5

A 4×4 grid of squares is coloured so that every row and every column contains exactly two black squares. How many black squares are there in the grid?

Answer: D

Each of the 4 rows contains exactly 2 black squares: 8 in total.

Full method - how and why

Count by rows - the column condition is extra information, not needed for the count.

- 1 There are 4 rows and each contains exactly 2 black squares.
why: Every black square lies in exactly one row, so rows partition them.
- 2 Total = $4 \times 2 = 8$.
why: No double counting.
- 3 Check with the columns: $4 \times 2 = 8$ as well - consistent, which is why such a colouring can exist.
why: If the two counts disagreed, no such grid would exist.

Where it goes wrong

- Adding rows and columns to get 16 - that counts every black square twice.
- Answering 4 by counting rows rather than squares.

Takeaway Count via one partition only; use the other as a consistency check, not as extra squares.

Question 11 SEQ-12 · Sequences and series · spec MM2.3

Which of the following geometric series is/are convergent?

I $1 + 0.9 + 0.81 + \dots$

II $3 - 3 + 3 - 3 + \dots$

III $100 + 10 + 1 + 0.1 + \dots$

Answer: F

A geometric series converges if and only if $|r| < 1$. I has $r = 0.9$, III has $r = 0.1$: both converge. II has $r = -1$ and does not converge.

Full method - how and why

Convergence of a geometric series depends only on $|r| < 1$. Compute r for each and apply the test.

1 I: $r = \frac{0.9}{1} = 0.9$, and $|0.9| < 1$: **converges**.

why: A ratio just below 1 still converges, though slowly.

2 II: $r = \frac{-3}{3} = -1$, and $|-1| = 1$ is not less than 1: **does not converge**.

why: The partial sums oscillate between 3 and 0 forever - they never settle, so there is no sum to infinity.

3 III: $r = \frac{10}{100} = 0.1$: **converges**.

why: Small ratio, rapid convergence.

4 I and III only.

why: Match the verdicts to the option list.

Where it goes wrong

- Thinking II converges to 0 or to 1.5 because the terms 'average out' - the partial sums must approach a single limit.
- Judging by the size of the terms rather than the ratio: III starts at 100 but converges fastest.

Takeaway Convergence is decided by $|r| < 1$ alone - not by the size of the first term or by whether the terms alternate.

Question 12 X2-10 · Circles and circle theorems · spec MM3.2

The circle C has centre (a, b) and radius r . P : ' $r > |b|$ '. Q : ' C crosses the x -axis at two distinct points'. Which one of the following is true?

Answer: A

The distance from the centre to the x -axis is $|b|$; the circle cuts the axis twice exactly when this is less than r .

Full method - how and why

Translate the geometry into a distance comparison: circle versus line depends on centre-to-line distance against the radius.

1 The perpendicular distance from the centre (a, b) to the x -axis is $|b|$.

why: The x -axis is $y = 0$, so the vertical distance is the size of the y -coordinate.

2 If $|b| < r$ the circle reaches across the axis and cuts it twice; if $|b| = r$ it touches once; if $|b| > r$ it misses entirely.

why: Three cases, decided by comparing distance with radius.

3 So 'two distinct points' happens exactly when $r > |b|$ - the two statements are equivalent.

why: Necessary **and** sufficient.

Where it goes wrong

- Writing the condition as $r > b$, which fails for a centre below the axis (b negative). The modulus matters.
- Including the tangent case $r = |b|$, which gives one point, not two.

Takeaway Circle versus line: compare the perpendicular distance from the centre with r - less gives two points, equal one, greater none.

Question 13 PRF-12 · Mathematical proof · spec Prf2

For a positive integer n it is known that: (i) if n is a multiple of 4 then n is not a multiple of 6; (ii) n is a multiple of 12 or n is a multiple of 4. What can be deduced?

Answer: A

If n were a multiple of 12 it would be a multiple of both 4 and 6, contradicting (i). So by (ii) n is a multiple of 4, and by (i) not a multiple of 6.

Full method - how and why

Deduction from two premises. Test the strongest-looking option against premise (i) first.

1 Suppose n were a multiple of 12. Then $12 = 4 \times 3$ and $12 = 6 \times 2$, so n is a multiple of both 4 and 6.

why: Exploring the tempting case first often resolves the whole question.

2 But (i) says a multiple of 4 is not a multiple of 6 - contradiction. So n is **not** a multiple of 12.

why: A supposition leading to a contradiction is ruled out.

3 Premise (ii) offers two alternatives and the first is now impossible, so n must be a multiple of 4.

why: Disjunctive syllogism: 'A or B' plus 'not A' gives B.

4 Then (i) applies: n is a multiple of 4 but not of 6.

why: That is the full deduction.

Where it goes wrong

- Concluding n is a multiple of 12 by picking the first branch of (ii) without testing it.
- Declaring the premises inconsistent - they are perfectly consistent (e.g. $n = 4$ or $n = 8$).
- Concluding nothing, missing that one branch of the 'or' is eliminated.

Takeaway With 'A or B', eliminate one branch and the other must hold; always test the branch that clashes with the other premise.

Question 14 FUN-12 · Functions and transformations · spec MM1.7

Let $f(x) = \sqrt{x-3}$ for $x \geq 3$ and $g(x) = x^2 + 3$. Which of the following statements is/are true?

- I $g(f(x)) = x$ for all $x \geq 3$
- II $f(g(x)) = x$ for all real x
- III $f(g(x)) = |x|$ for all real x

Answer: F

I: $g(f(x)) = (\sqrt{x-3})^2 + 3 = x$ for $x \geq 3$: true. $f(g(x)) = \sqrt{x^2 + 3} = |x|$, so II fails (e.g. $x = -1$) and III is true.

Full method - how and why

Two functions that almost invert each other. The asymmetry comes from $\sqrt{x^2} = |x|$ and from the restricted domain.

1 I: $g(f(x)) = (\sqrt{x-3})^2 + 3 = (x-3) + 3 = x$ for $x \geq 3$. **True.**

why: Squaring a square root recovers the inside, provided the inside is non-negative - guaranteed by the stated domain $x \geq 3$.

2 Compute the other order: $f(g(x)) = \sqrt{(x^2 + 3) - 3} = \sqrt{x^2} = |x|$.

why: Do the algebra once and read both remaining statements off it.

3 II claims this equals x for all real x : false at $x = -1$, where $|x| = 1 \neq -1$. **False.**

why: The square root returns the non-negative value, so it cannot reproduce a negative input.

4 III says it equals $|x|$, which is exactly what we found. **True.**

why: Same computation, correct conclusion.

5 I and III only.

why: Match to the options.

Where it goes wrong

- Assuming $f(g(x)) = x$ because f and g 'look like' inverses - inverses require matching domains and ranges.
- Writing $\sqrt{x^2} = x$.

Takeaway $f(g(x)) = x$ and $g(f(x)) = x$ are different statements; $\sqrt{x^2} = |x|$ is what breaks the symmetry here.

Question 15 X2-11 · Functions and transformations · spec MM8.2

Let $f(x) = x^2$ for all real x , and let g be the function whose graph is obtained by translating the graph of $y = f(x)$ by 2 units in the positive x -direction and then reflecting in the y -axis. Which of the following statements is/are true?

- I** $g(x) = (x + 2)^2$
- II** $g(x) = (x - 2)^2$
- III** The graph of g has its vertex at $(-2, 0)$

Answer: F

Translation gives $(x - 2)^2$; reflecting in the y -axis replaces x by $-x$: $(-x - 2)^2 = (x + 2)^2$, with vertex at $(-2, 0)$.

Full method - how and why

Composite transformations, applied in the stated order. Each transformation acts on the function you currently have.

- 1** Translate $y = x^2$ by 2 in the positive x -direction: replace x by $x - 2$, giving $y = (x - 2)^2$.
why: A shift **right** subtracts inside the bracket - counter-intuitive but standard.
- 2** Now reflect in the y -axis: replace x by $-x$, giving $y = (-x - 2)^2$.
why: Reflection in the y -axis is the substitution $x \rightarrow -x$.
- 3** Simplify: $(-x - 2)^2 = (-(x + 2))^2 = (x + 2)^2$. So $g(x) = (x + 2)^2$ - **I is true, II is false.**
why: Squaring removes the overall minus sign.
- 4** III: $(x + 2)^2$ has its vertex where $x + 2 = 0$, i.e. at $(-2, 0)$. **True.**
why: Consistent: the vertex moved right to $(2, 0)$, then reflected to $(-2, 0)$.
- 5** So I and III.
why: Match to the list.

Where it goes wrong

- Doing the transformations in the wrong order - reflecting first then translating gives $(x - 2)^2$ instead.
- Shifting right by adding 2 inside the bracket.
- Forgetting to simplify $(-x - 2)^2$, and so failing to recognise it as $(x + 2)^2$.

Takeaway Right by a : $x \rightarrow x - a$. Reflect in the y -axis: $x \rightarrow -x$. Apply them in the order given, tracking the vertex as a check.

Question 16 RSN-17 · Chains of reasoning · spec Prf5

Alice, Ben and Cara each make one statement. Alice: 'Ben is lying.' Ben: 'Cara is lying.' Cara: 'Alice and Ben are both lying.' Exactly one of them is telling the truth. Who is it?

Answer: A

If Alice is truthful then Ben lies, so Cara is truthful: two truth-tellers, impossible. If Cara is truthful then Alice and Ben both lie; but Alice lying means Ben is truthful: contradiction. So Ben is the truth-teller: Cara lies (indeed Ben is not lying) and Alice lies (Ben is not lying), which is consistent.

Full method - how and why

A truth-teller puzzle. Assume each person in turn is the sole truth-teller and check the whole scenario for consistency.

- 1 Suppose **Alice** is truthful. Then Ben lies, so 'Cara is lying' is false, so Cara is truthful. That makes two truth-tellers - contradiction.
why: Exactly one truth-teller is the constraint; two breaks it.
- 2 Suppose **Cara** is truthful. Then Alice and Ben both lie. But Alice lying means 'Ben is lying' is false, so Ben is truthful - contradiction.
why: Cara's own statement collapses under its consequences.
- 3 Suppose **Ben** is truthful. Then Cara lies, so Ben is not lying ✓ consistent. Alice says 'Ben is lying', which is false, so Alice lies ✓. Cara says both Alice and Ben lie - false, since Ben is truthful ✓.
why: Every statement's truth value matches the assumed roles, with exactly one truth-teller.
- 4 So Ben is the truth-teller.
why: Two cases eliminated, one consistent - the answer is determined.

Where it goes wrong

- Stopping at the first scenario that seems to work without checking every statement.
- Declaring the situation impossible after two failed cases instead of testing the third.
- Forgetting that a liar's statement must come out **false**, which is itself information.

Takeaway Test each candidate exhaustively: a consistent assignment must satisfy every statement **and** the count of truth-tellers.

Question 17 ERR-07 · Identifying errors in proofs · spec Err2

A student writes the following argument.

I $0.1 > 0.01$.

II Taking logarithms to base 10 of both sides: $-1 > -2$.

III Multiplying both sides by $\log_{10} \frac{1}{2}$, which is negative, reverses the inequality:
 $-\log_{10} \frac{1}{2} < -2\log_{10} \frac{1}{2}$.

Which of the following is correct?

Answer: A

Logarithms are increasing functions, so line II is valid; $\log_{10} \frac{1}{2} < 0$, so reversing the inequality in line III is correct.

Full method - how and why

Sometimes the argument is simply valid. Check each step against the rules rather than assuming a trap.

1 Line I: $0.1 > 0.01$ ✓.

why: $\frac{1}{10} > \frac{1}{100}$.

2 Line II: $\log_{10}$ is an **increasing** function, so it preserves the inequality: $\log_{10} 0.1 = -1$ and $\log_{10} 0.01 = -2$, and indeed $-1 > -2$ ✓.

why: Applying an increasing function to both sides keeps the direction; $\log_{10} 10^{-1} = -1$.

3 Line III: $\log_{10} \frac{1}{2} < 0$ because $\frac{1}{2} < 1$. Multiplying an inequality by a negative number reverses it, which is exactly what is written ✓.

why: $-1 > -2$ becomes $-1 \times c < -2 \times c$ for $c < 0$.

4 So all three lines are correct.

why: The question is testing whether you can certify a valid argument, not only spot faults.

Where it goes wrong

- Assuming logarithms reverse inequalities - they do not; $\log$ is increasing on its domain.
- Thinking $\log_{10} \frac{1}{2}$ is positive; logs of numbers below 1 are negative.
- Misreading $\log_{10} 0.1$ as 1 rather than -1 .
- Hunting for an error simply because the question implies one might exist.

Takeaway Increasing functions preserve inequalities; multiplying by a negative reverses them. $\log_b x < 0$ exactly when $0 < x < 1$ (for $b > 1$).

Question 18 X2-28 · Statistics · spec M6.3

A set of five distinct positive integers has median 10 and mean 10. Which of the following statements must be true?

- I The largest value is at least 12.
- II The smallest value is at most 8.
- III The range is at least 4.

Answer: H

The two values above the median are distinct integers greater than 10, so at least 11 and 12 (I); similarly the two below are at most 9 and 8 (II); hence the range is at least $12 - 8 = 4$ (III).

Full method - how and why

Extremal reasoning with **distinct integers**: the distinctness forces gaps of at least 1 on each side of the median.

1 Order the five values. The median (third value) is 10.

why: Five values means the middle one is the third.

2 The two above are distinct integers greater than 10, so at least 11 and 12. Hence the largest is ≥ 12 . **I true.**

why: Distinct integers above 10 must be 11 or more, and two of them cannot both be 11.

3 Symmetrically, the two below are at most 9 and 8, so the smallest is ≤ 8 . **II true.**

why: Same argument downwards.

4 Range $\geq 12 - 8 = 4$. **III true.**

why: Combining the two bounds. (The mean being 10 is consistent - e.g. 8, 9, 10, 11, 12 - but is not needed for any of the three claims.)

5 So all three.

why: Match to the list.

Where it goes wrong

- Forgetting 'distinct' and allowing two values equal to 11, which would break I.
- Assuming the mean condition must be used; here it is extra information.

Takeaway Distinct integers force minimum gaps: k values above a median m reach at least $m + k$.

Question 19 PRF-18 · Mathematical proof · spec Prf2

Every element of a set S of integers is either a multiple of 3 or a multiple of 5. Every multiple of 3 in S is negative. S contains the number 10. Which of the following statements must be true?

- I S contains a positive multiple of 5.
- II S contains no positive multiple of 3.
- III S contains a negative number.

Answer: E

$10 \in S$ is a positive multiple of 5 (I). Multiples of 3 in S are negative, so none is positive (II). III fails: $S = \{10\}$ satisfies everything.

Full method - how and why

'Must be true' with a concrete object given. Some claims follow from the given element; others need a counterexample set.

- 1 I: $10 \in S$ and $10 = 2 \times 5$ is a positive multiple of 5. **Must be true.**
why: The element is handed to you; nothing more is needed.
- 2 II: the second condition says every multiple of 3 in S is negative, so none can be positive. **Must be true.**
why: Note this is vacuously fine even if S contains no multiples of 3 at all.
- 3 III: must S contain a negative number? Try $S = \{10\}$: every element is a multiple of 3 or 5 ✓, there are no multiples of 3 to be negative ✓, and $10 \in S$ ✓. No negative number. **Not necessarily true.**
why: A single valid S with no negative element refutes 'must be true'.
- 4 So I and II.
why: Match to the list.

Where it goes wrong

- Reading 'every multiple of 3 in S is negative' as promising that S **has** a multiple of 3 - a universal statement can be true with nothing to apply it to.
- Confusing 'could be true' with 'must be true' for III.

Takeaway A conditional statement about members of a set is vacuously true when no member qualifies; to refute 'must be true', build a legal example that fails it.

Question 20 IND-12 · Indices and surds · spec MM1.1

For how many integer values of n with $1 \leq n \leq 100$ is $\sqrt{n} + \sqrt{n+1}$ rational?

Answer: D

For the sum to be rational, both $\sqrt{n}$ and $\sqrt{n+1}$ would need to be rational (if one is irrational, $\sqrt{n} + \sqrt{n+1}$ squared gives $2n+1 + 2\sqrt{n(n+1)}$ and $n(n+1)$ is never a perfect square for $n \geq 1$ since $n^2 < n(n+1) < (n+1)^2$). No two consecutive positive integers are both squares, so the answer is 0.

Full method - how and why

A 'for how many' question that is really about irrationality. The answer is 0, and seeing why needs one sharp observation about consecutive integers.

- 1 Suppose $\sqrt{n} + \sqrt{n+1} = r$ is rational and square it: $2n+1 + 2\sqrt{n(n+1)} = r^2$.
why: Squaring is the standard way to get at a sum of surds; the rational parts separate from the single remaining surd.
- 2 Rearrange: $\sqrt{n(n+1)} = \frac{r^2 - 2n - 1}{2}$, so $n(n+1)$ would have to be a perfect square.
why: A rational square root of an integer is an integer, so $n(n+1)$ must be a perfect square.
- 3 But $n^2 < n(n+1) < (n+1)^2$ for every $n \geq 1$, so $n(n+1)$ lies strictly between consecutive squares.
why: There is no integer square strictly between n^2 and $(n+1)^2$ - this is the contradiction.
- 4 Hence no n in $1 \leq n \leq 100$ works: the answer is 0.
why: The bound 100 is a red herring; the argument rules out every positive integer.

Where it goes wrong

- Testing $n = 3$ ($\sqrt{3} + 2$) and concluding 'sometimes it works' because one of the two roots is nice.
- Answering 9 or 10 by counting perfect squares up to 100 - that would answer a different question.

Takeaway $n(n+1)$ is never a perfect square for $n \geq 1$, because it is squeezed strictly between n^2 and $(n+1)^2$.